

THE INDUSTRIAL REVOLUTION PRE-CONDITIONS

ALFANI COLLI, CH. 5

Economic history

A.y. 2025/2026

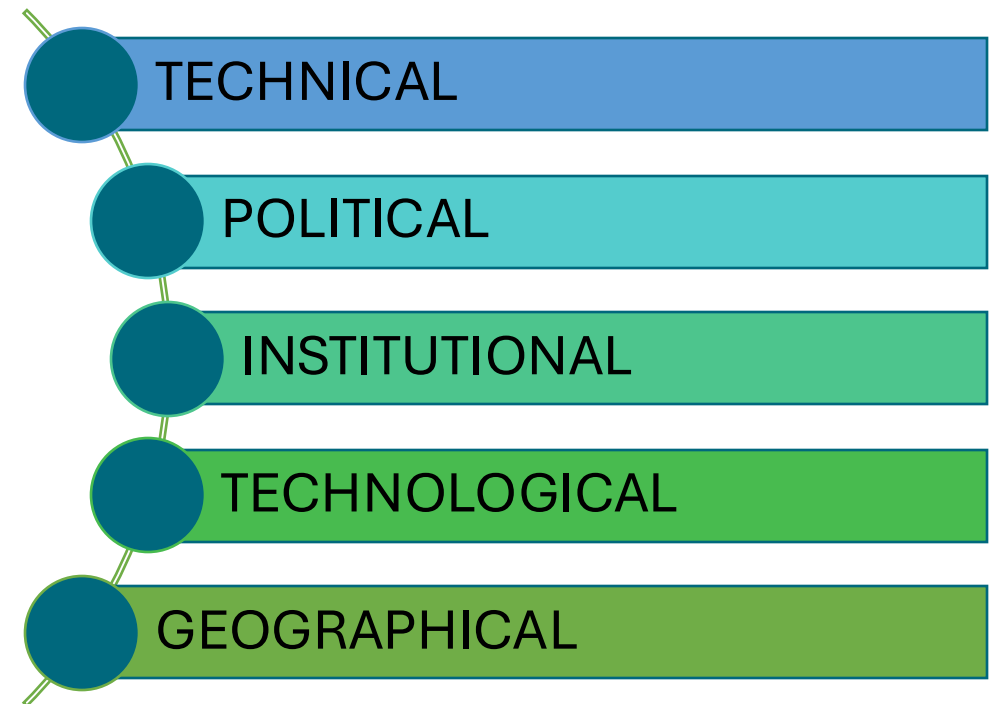
Sonia Schifano

IN THIS CLASS

- Preconditions
- Factors of the IR
 - Market oriented economy
 - The role of the State
 - Resources
- Technology
- Sectors
 - Textile
 - Energy
- Why Britain?
- Obstacles: advantage or disadvantage?

BREAKING POINT WRT PAST

- In the attempt to increase productivity, what changed from the past is the shift
 - from specialisation and trade
TO
 - investment and technological progress
- The opening up of the Atlantic trade routes allowed the increase of the share of non-agro population
- As population grows
 - People need to consume food to survive.
 - People can only start producing non-agricultural goods, if farmers produce a surplus that can be traded for these goods.



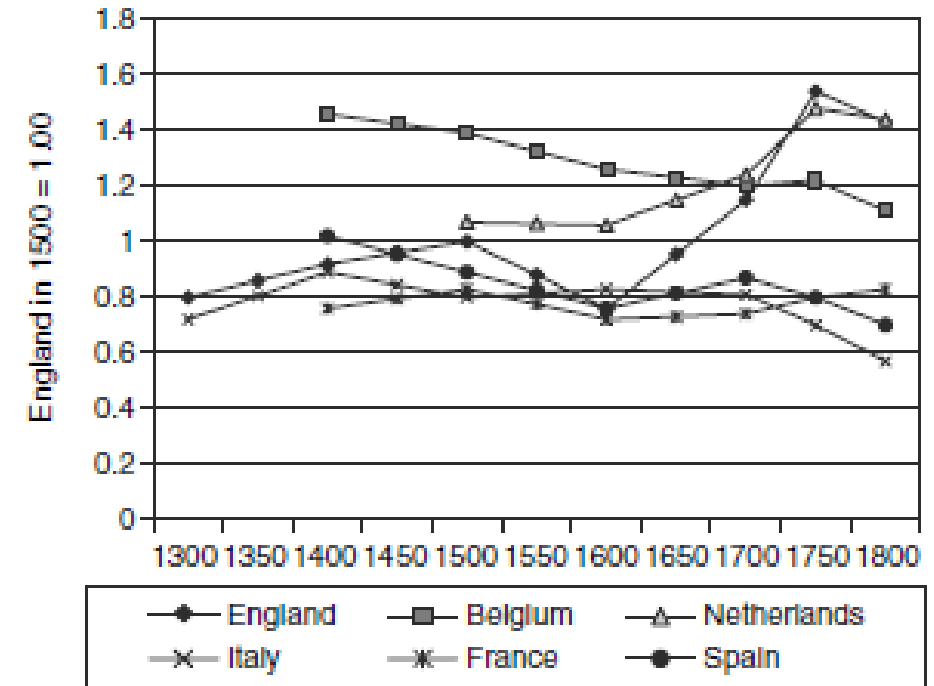
THE AGRICULTURAL REVOLUTION 1400-1700

THE HIGH WAGES ECONOMY ALLOWED TO SUSTAIN THE DEMAND OF GOODS

- Manufactured goods → market incentive
- Better food → dairy and meat → shifted grain production to animal grazing and livestock

More draft animals → more draft power and more manure → higher labour productivity and yields → even higher wages

RESULT: **INCREASE IN OUTPUT PER WORKER**



Agricultural labour productivity

INCREASE IN OUTPUT PER WORKER

1. How did farmers increase productivity?

- By increasing capital intensity (draft animals)
- By increasing land area per farm worker
- By improving techniques and soil quality

	1600	1700	1750	1800
England	53,1	80,4	107,7	100,0
Belgium	88,1	83,9	85,3	77,6
Netherlands	74,1	86,7	103,5	100,7
France	50,3	51,7	55,9	58,0
Italy	58,0	56,6	49,0	39,9
Spain	53,1	60,8	55,9	49,0
Germany	39,9	37,8	39,2	46,9
Austria	39,9	51,7	69,9	51,5
Poland	54,5	65,7	65,0	74,8

Allen (2000) Agricultural labour productivity in Europe (England 1800 = 100).

MARKET ORIENTED ECONOMY!

FROM OPEN FIELDS TO ENCLOSURES

- Before the Black Death, the dominant form of land tenure in Europe was the open-field system:
 - Fields divided up between villagers into narrow strips
- As demand for grain fell and demand for English wool in Flanders increased, English lords enclosed open fields.
 - **1500:** 45% of land enclosed
 - **1750:** nearly two-thirds of land enclosed
 - **1760-1830:** Parliamentary enclosures
- In 18th century, English agriculture dominated by consolidated landholdings rented to capitalist farmers.

FROM OPEN FIELDS TO ENCLOSURES

- English historians have explained the rise of English agriculture with the enclosure of open fields.
- Inefficient open fields
 - Villagers forced to cooperate → hard to innovate
 - Peasants had separate and often far away holdings
 - Lands rotated among villagers → no incentive to invest
- Efficient enclosed fields
 - Large consolidated holdings → economies of scale
 - Tenant farmers invest in capital to economise on labour
 - Long lease → tenants interested in improving or maintaining yields



(What about inequality?)

FROM OPEN FIELDS TO ENCLOSURES

- Technology
 - Adoption of new crops and rotations
- Land use
 - Shift from arable to grassland
- Crop yields
 - Geographical differences
 - Type of crops
- Labour productivity

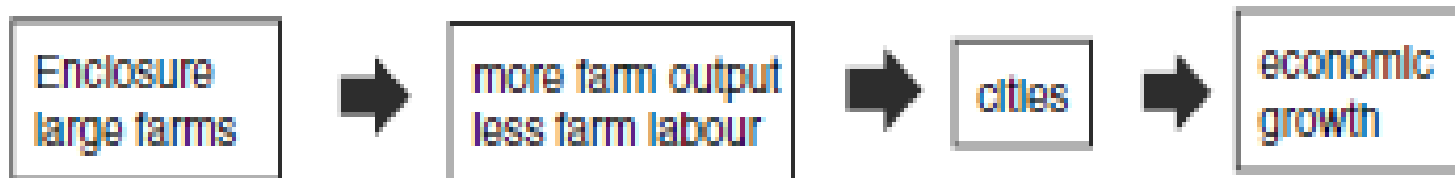
Table 3.1 *Crop yields and enclosure, c. 1800*

	Open bushels per acre	Enclosed bushel per acre	Enclosed relative to open	Gain relative to yield increase from middle ages to nineteenth- century enclosures
Heavy arable district				
Wheat	19.7	20.2	2.2	5.3
Barley	26.5	31.8	20.0	35.3
Oats	23.5	33.0	40.4	44.6
Beans	18.8	22.2	18.1	27.9
Average	21.2	24.1	14.7	23.8
Light arable district				
Wheat	20.0	19.7	-1.5	““““
Barley	27.0	29.3	8.5	18.4
Oats	26.5	32.5	22.6	28.8
Beans	19.9	18.1	-9.0	““““
Average	23.4	24.7	5.6	10.9
Pasture district				
Wheat	20.9	21.9	4.8	8.9
Barley	28.0	32.2	15.0	27.3
Oats	36.9	38.1	3.3	4.5
Beans	22.4	23.4	4.5	7.5
Average	24.7	26.7	8.1	14.2

Source: Allen (1992, p. 136).

THE PROCESS 1.0

- Traditional view
 - Enclosures reduced the demand for farm labour and increased the food surplus, which fuelled urban growth.
 - Larger urban population → greater specialisation & trade



THE PROCESS 2.0

Allen offers an alternative explanation to the traditional view. Innovation in English agriculture was the consequence, not the cause, of urban growth.

- The impact of city growth
 - Agricultural wages fell behind urban wages
 - Fast urban growth increased the relative price of food
- Consequences for farmers
 - High urban wages pulled farmers into the city
 - High food prices lured capital into agriculture
 - Enclosures: consolidated land holdings close to cities
- Impact on productivity
 - Increased land-labour ratio
 - Induced innovation: farmers had to innovate to keep up with rising urban wages

PRODUCTIVITY GROWTH

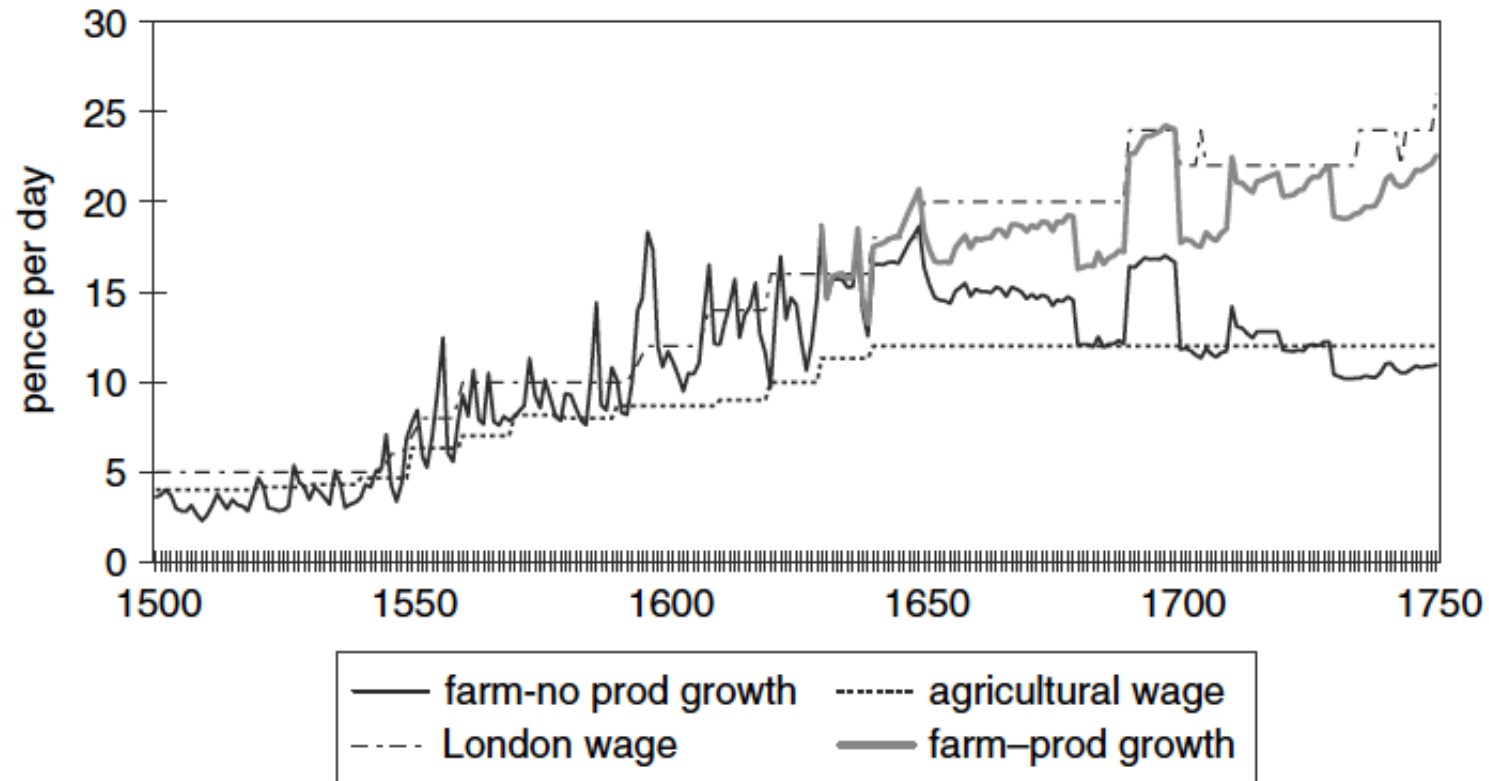


Figure 3.4 Alternative farm incomes with and without productivity growth

CAN WE ESTABLISH A CAUSAL RELATIONSHIP?

Beware that the interpretation of the role of agriculture in the economic development process is not unanimous...



Which one is correct? The old one (↑) or the more recent one (↓)?

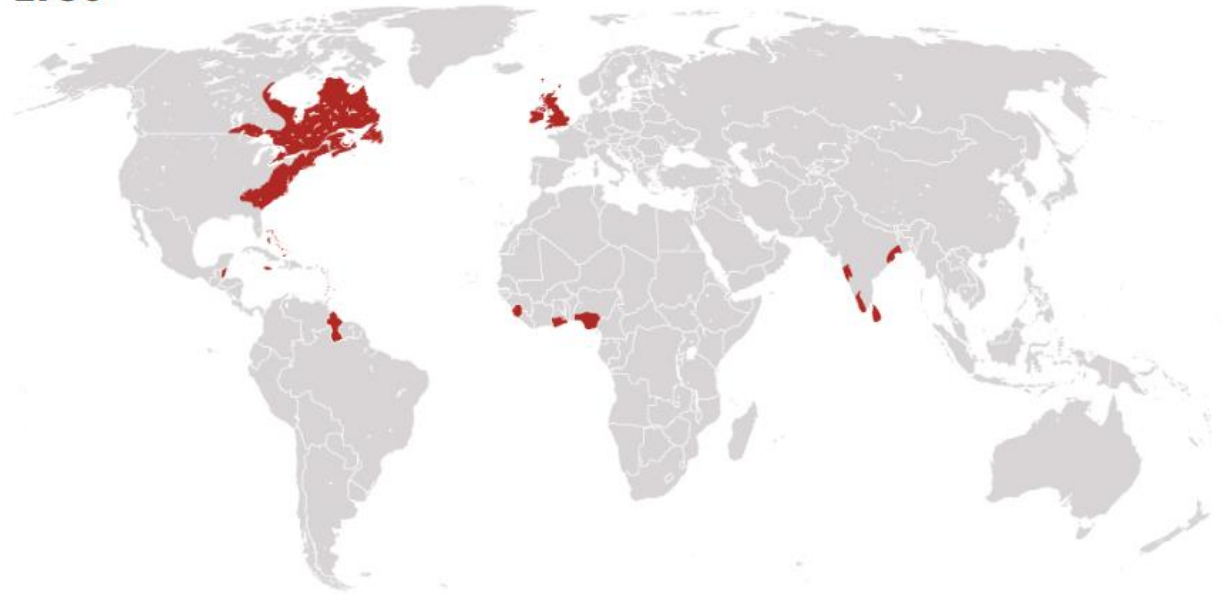


THE ROLE OF THE STATE

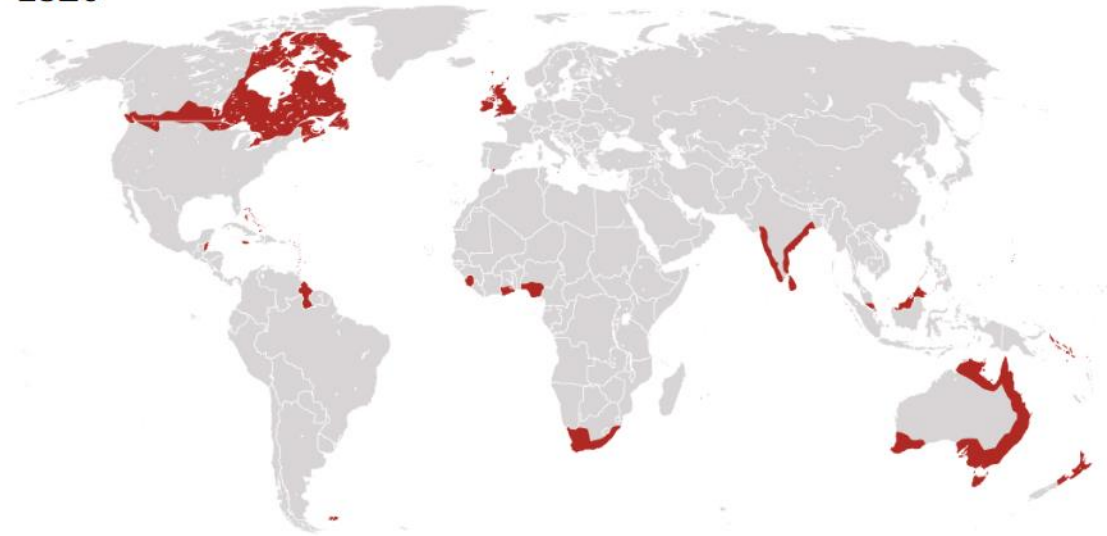
1680



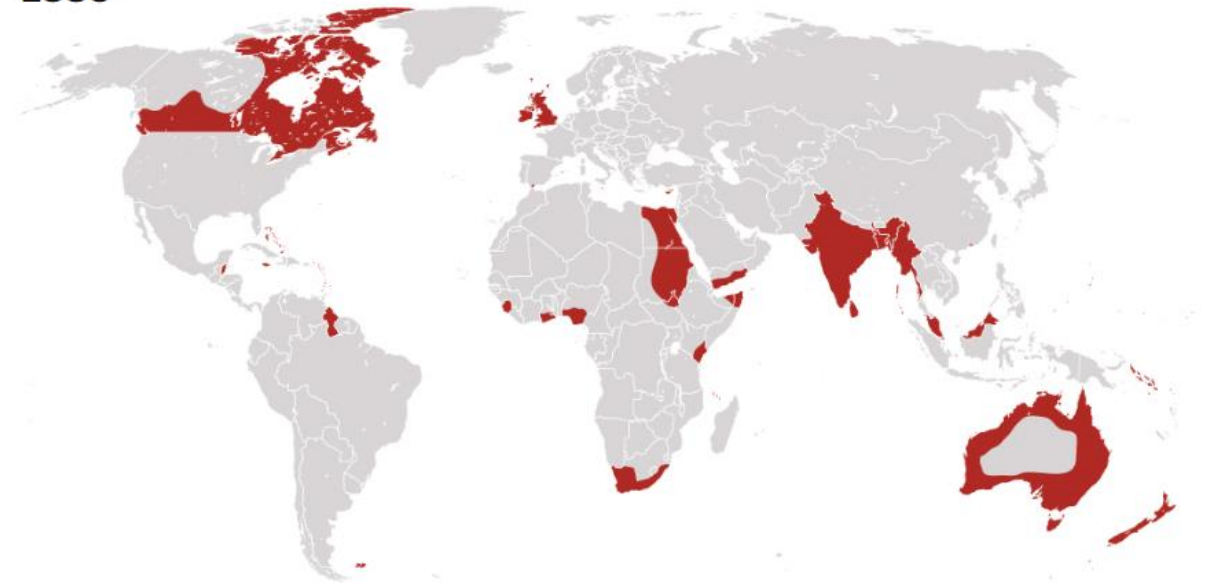
1750



1820



1885



MERCANTILISM & EMPIRE

With **mercantilism, or economic nationalism**, the aim of the state is (like the good merchant) to earn as much as possible by exporting as much, but importing as little as possible

→ **WEALTH ACCUMULATION**

- Mercantilist policies base on:
 - **strengthening trade with the colonies** → increasing access to resources and the British market
 - **dominance over the seas** to ensure high levels of exports
 - they were also intended to strengthen **domestic production**.
- Mercantilist policies were implemented to **strengthen the state and to “commercially” fight the Dutch**
 - both financially and institutionally

MERCANTILISM & EMPIRE

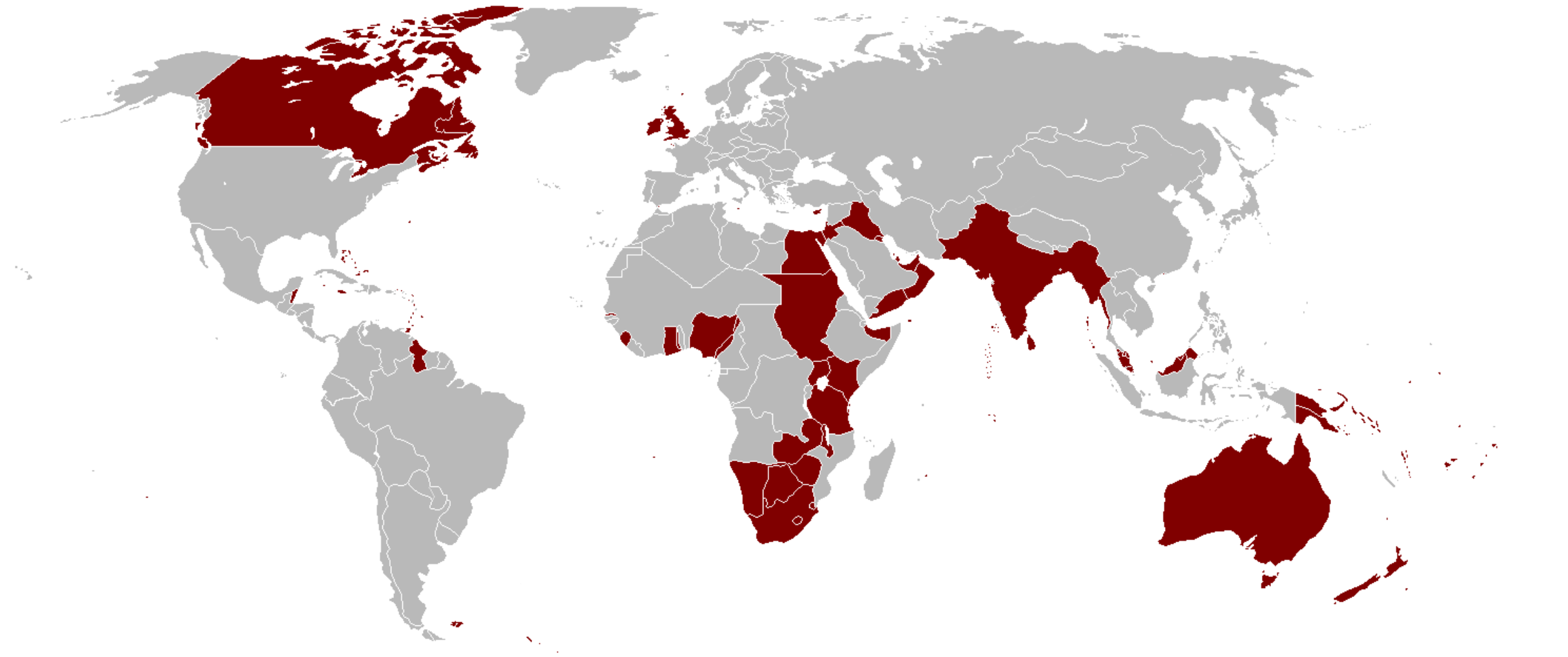
Among the most significant mercantilist policies are the **Navigation Acts**, a symbol of the Old Colonial System. There are three: **1651, 1660, 1663**.

- **Goals:** Strengthen English shipping, secure control of colonial trade, and weaken rivals (Dutch & French).
- **Main rules:**
 - Only **British ships** could transport goods to and from England and its colonies.
 - Salted fish had to travel **exclusively on British vessels**.
 - Restriction on certain colonial exports, ensuring they were primarily sold to **Britain**.
 - Crews and shipowners had to be English (or colonial subjects).
- **Economic effects:**
 - Created a **protected English shipping industry**.
 - Raised customs duties, even for trade between colonies.
- **Political & military impact:**
 - Triggered a century of conflict with the Dutch and French.
 - First **Anglo-Dutch War broke out in 1652**.

MERCANTILISM & EMPIRE

- The third navigation act, the **Staple Act of 1663**, required instead that the English colonies had to obtain their supplies only from Great Britain and that certain goods (sugar, tobacco, cotton, etc.) had first to be brought to the mother country and then, eventually, re-exported.
- These measures led to furious clashes with the Dutch and French that lasted for at least a century (approximately from 1652 to 1763).
- The **costs of the wars were very high** and to avoid raising the tax burden too much in Britain, parliament tried to raise taxes in the colonies.
- Hardest hit were the inhabitants of the 13 American colonies, who proclaimed their independence with the cry of “*no taxation without representation*”.

The empire at its peak (1921)



INSTITUTIONS

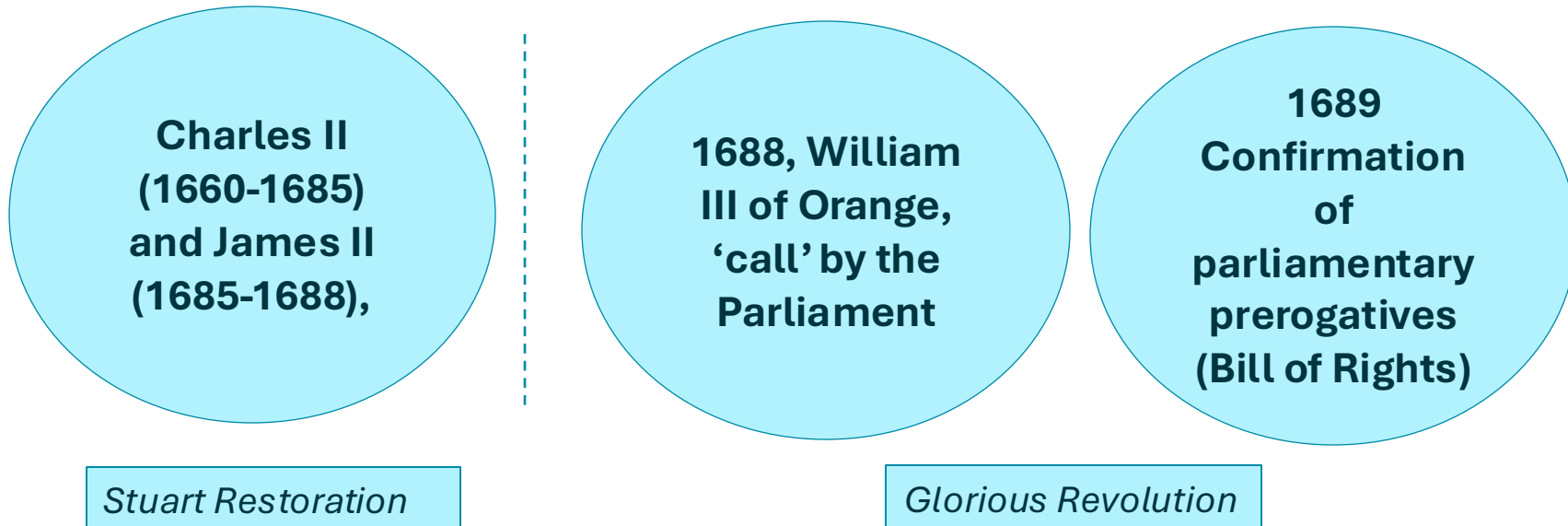
- The Civil War (1642-1649) broke out after several decades of political tension between the Crown and the Parliament. It ended with the death sentence of Charles I (1649).
- Main source of tension: the king's right to impose taxes on his subjects without parliamentary approval.
- Republican experience → Commonwealth (1649-1660)
- The 'Glorious Revolution' (1688-89) made the English political system more 'inclusive', i.e., more open to the needs of the business community by strengthening property rights, perfecting financial systems, and removing barriers to the expansion of industry and trade.



THE CAUSES OF THE BRITISH IR: INSTITUTIONS

The British Parliament began to regulate, on a macro-economic level:

- trade policy → e.g Navigation Acts
- fiscal policy
- monetary policy
- labour market → abolition of Guilds



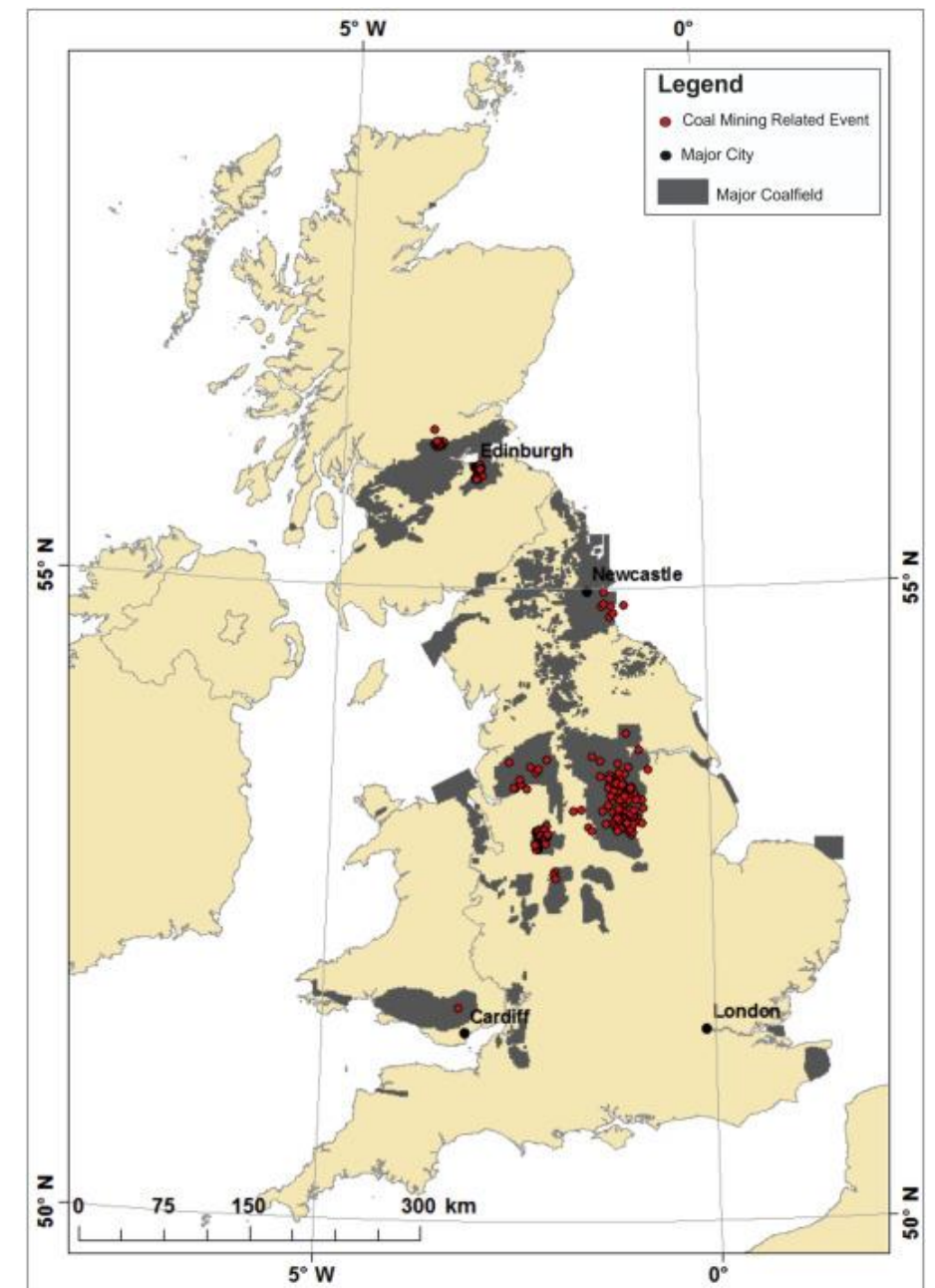
RESOURCES

COAL

- Historians for long attributed the British industrial revolution to the early development of coal mining in England.
- ‘Britain was first because Britain had coal’
 - By 18th century, Britain produced most of the world’s coal
 - Southeast Belgium the only other region of significant production
 - Rest of the world continued to use traditional energy sources
- The ‘timber crisis’ theory (Nef 1932)
 - By 17th century, England had cut down its forests for heating fuel and material for shipbuilding.
 - Timber shortage → wood and charcoal had to be replaced by coal
 - Coal mining increased 66-fold between 1560 and 1800

IMPORTANCE OF COAL

- Shift from 'organic' to 'mineral-fuel' economy
- First modern industry
 - Geographic concentration (northeast)
 - Large scale production
 - High investment (capital) requirements
 - Strong industry linkages (finance, insurance, shipping)
- Technological spinoffs
 - Steam engine
 - Mass production of iron and steel
 - Railways



THE CHEAP ENERGY ECONOMY

- Early development of coal made Britain the world's cheap energy economy, which empowered the industrial revolution.
- Demand mattered, not supply
 - Several nations possessed large reserves of coal, but only Britain developed a large coal industry before 1800.
 - Growth of coal industry driven by growth of London
 1. Booming demand for cheap heating fuel
 2. Innovation in heating technology driven by housing boom
- Spillover effects:
 - Once coal-burning house was invented, it spread to smaller towns, giving rise to coal industry in western Britain and Scotland.
 - Urban growth in the periphery created further demand for fuel and thus spurred the meteoric rise of coal mining.

THE CHEAP ENERGY ECONOMY

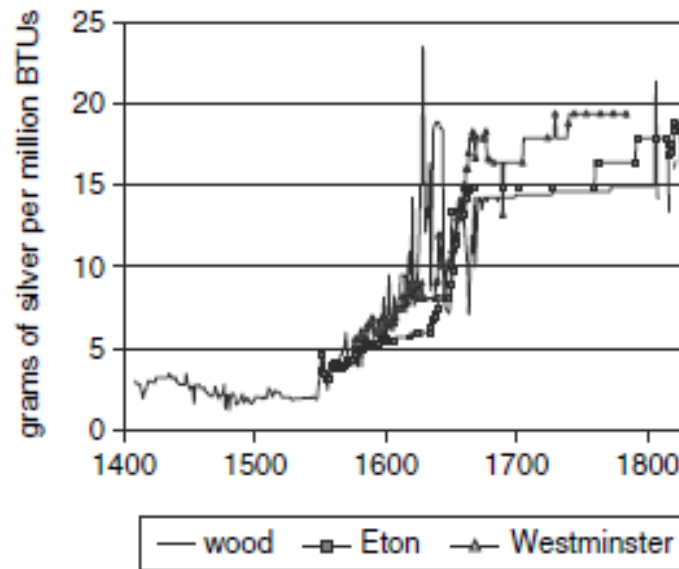


Figure 4.2 Prices of wood fuels in London

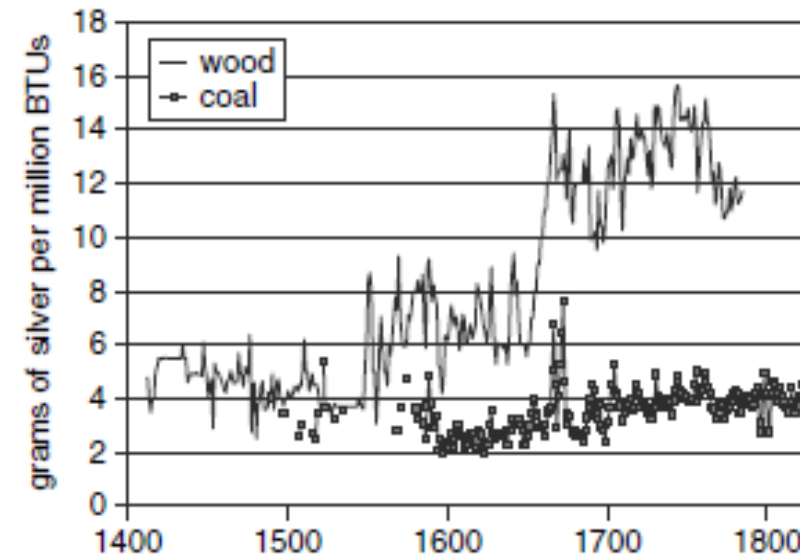


Figure 4.3 Real prices of wood and coal in London

Source: Allen (2009), 'Cheap energy economy'

TRANSPORTS

In the process of economic growth, the **transportation revolution** was a fundamental step.

Transport innovations was important because:

- they created a more **connected internal market**
 - connecting production and consumption centres
- they lowered **transaction costs** (-40%)
- they lowered **travel times** (-60%)
- they brought a progressive **convergence of prices**
- they favoured a more efficient regional allocation of factors (**specialisation**)

TRANSPORTS

Toll roads (*turnpike roads*) are built according to a new model:

- construction by turnpike trusts authorised by parliament
- financing through debt
- debt repayment through tolls

This model was one the most evident results of the *Bubble Act* of 1720, which prohibited the incorporation of joint stock companies without parliamentary approval (too many speculative bubbles in years had ended up upsetting policy makers...).

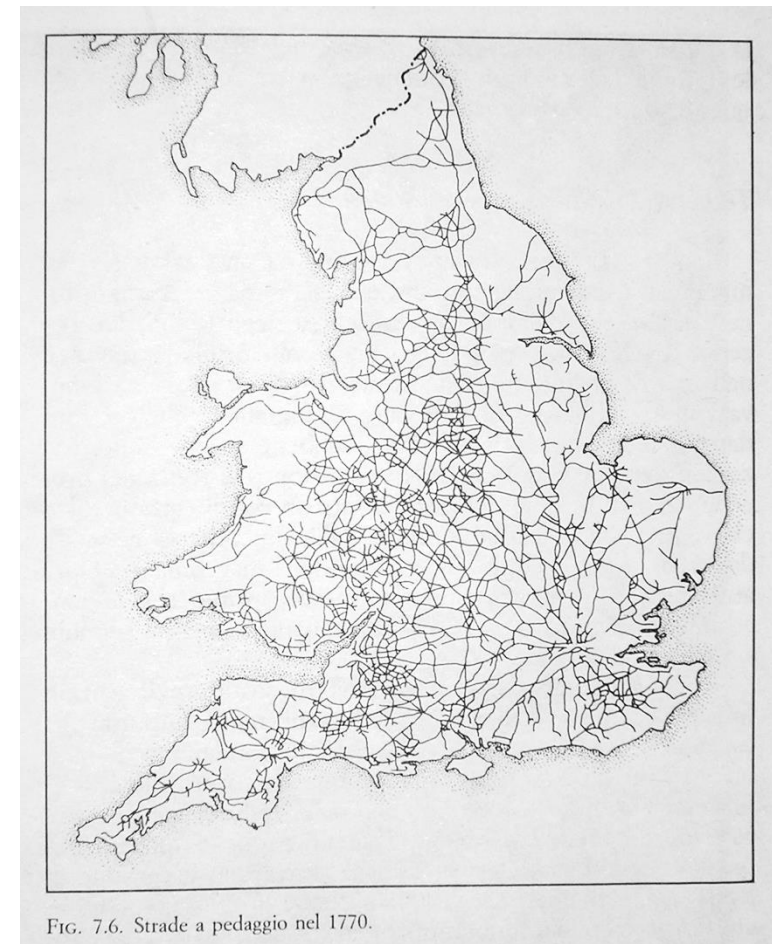
Moreover, we do not have to forget the fundamental role played by internal canals.

TRANSPORTS

Internal canals



Turnpike roads



WHAT WAS A CLEAR RUPTURE WRT THE PAST?

THE BREAKING POINT

The industrially more developed country
presents to the less
developed country a picture of the latter's
future.
Karl Marx, *Capital*, vol. I, preface

The term revolution indicates the **radical nature** of economic and social change rather than the speed of change.

- ▶ What has changed compared to the past?
 - Continuous population growth
 - GDP and GDP per capita growth
 - Increase in productivity
 - Increase in TFP (total factor productivity)
 - **Structural change** in the economy: decline of the agricultural sector
 - Social and cultural change (commoditization of land, capital, labour)
 - Radical change of the mindset
 - Formation of an increasingly global market

- ▶ According to Simon Kuznets (1966) this brought Europe towards “modern economic growth”.

EXAMPLE IN NUMBER: GDP PER CAPITA

Comparing GDP p/c growth

Table 2–22a. Levels of GDP Per Capita in European Colonial Powers and Former Colonies, 1500–1998
(1990 international dollars)

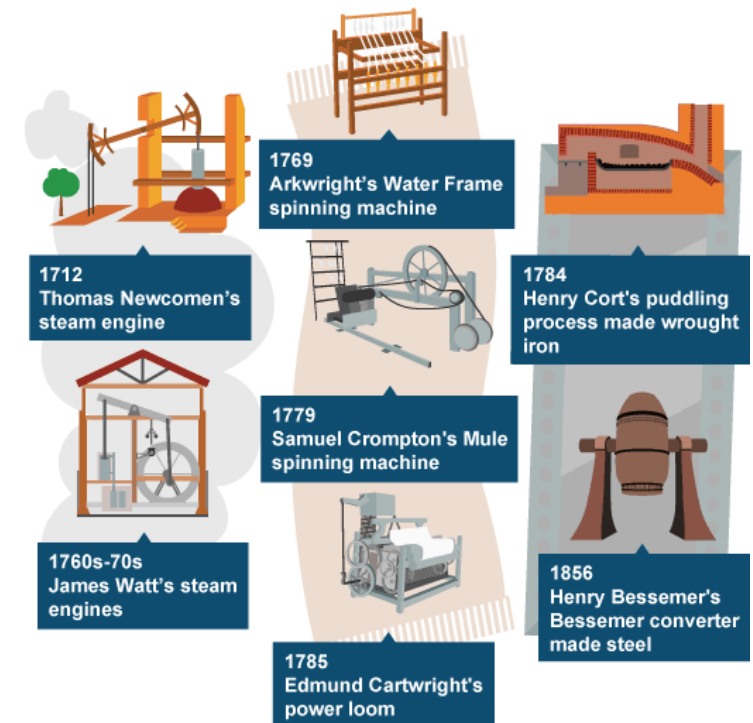
	1500	1700	1820	1913	1950	1998
Britain ^a	762	1 405	2 121	5 150	6 907	18 714
France	727	986	1 230	3 485	5 270	19 558
Italy	1 100	1 100	1 117	2 564	3 502	17 759
Netherlands	754	2 110	1 821	4 049	5 996	20 224
Portugal	632	854	963	1 244	2 069	12 929
Spain	698	900	1 063	2 255	2 397	14 227
China	600	600	600	552	439	3 117
India	550	550	533	673	619	1 746
Indonesia	565	580	612	904	840	3 070
Brazil	400	560	646	811	1 672	5 459
Mexico	425	568	759	1 732	2 365	6 655
United States	400	527	1 257	5 301	9 561	27 331
Ireland ^b	526	715	880	2 736	3 446	18 183

How was all this possible!?

TECHNOLOGY

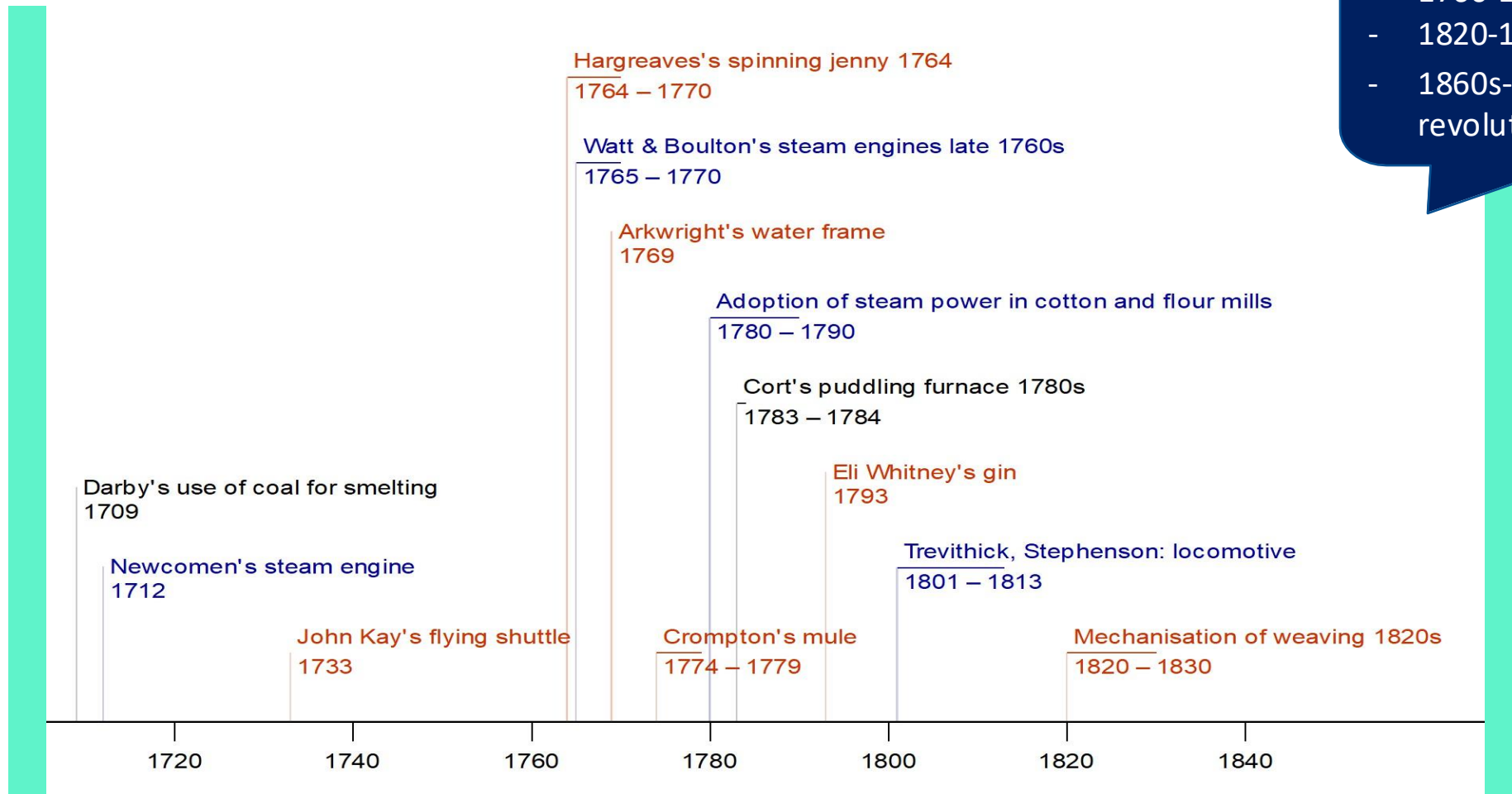
► New technologies:

1. allow to produce at lower costs than in the past
2. allow to achieve productivity growth in the long run
3. generate savings in the use of resources
4. progressively improve the quality of the goods



TECHNOLOGY THAT MATTERS

- 1760-1820: macro-innovations
- 1820-1850s: micro-innovations
- 1860s-: second industrial revolution



WHAT KIND OF TECHNOLOGICAL INNOVATION

“About 1760 a wave of gadgets swept through England” (Ashton, 1955)

- Process of innovation
 1. **Macro inventions:** entire new technologies that set in motion new technical advances and generate leaps in productivity
 2. **Micro inventions:** incremental improvements on existing technologies that enhance their application and increase their efficiency
- Nature of innovation
 - ✧ **Inspiration:** scientific discovery and new knowledge may be culturally determined or matter or ‘luck’.
 - ✧ **Perspiration:** innovation requires research and development that is costly and thus depends on economic incentives.

MICRO-INVENTIONS

- Neutral (not biased) technical change
 - **Micro inventions** are typically **factor neutral**: increase the productivity of all factors proportionately
 - **Reverse factor bias**: when improvements save on the input that the macro invention used most intensively (e.g. coal in steam engines)
- Local learning
 - Innovation takes place in the context of one industry, where inventors are already users of the technology they aim at perfecting.
 - Any reduction in cost improves profitability → no factor bias
- Collective invention
 - Information sharing makes micro-inventions cheap, reducing the need for financing and patents.

THE SLOW GROWTH PARADOX

Table 6.1 TFP growth and new and old estimates of national product growth in Britain during the Industrial Revolution. Per cent per year

	TFP	Revised estimates		Dean & Cole	
		National product	Per head	National product	Per head
1700–60		0.69	0.31	0.66	0.45
1760–80	0.14*	0.70	0.01	0.65	–0.04
1780–1801	0.14*	1.32	0.35	2.06	1.08
1801–31	0.41	1.97	0.52	3.06	1.61

Sources: N. F. R. Crafts, *British Economic Growth during the Industrial Revolution* (Oxford: Clarendon Press, 1985), and for TFP estimates based on Crafts' and Harley's revisions, J. Mokyr, 'Accounting for the Industrial Revolution', in R. Floud and P. Johnson (eds.), *The Cambridge Economic History of Modern Britain* (Cambridge University Press, 2004), pp. 1–27.

* The TFP growth of 0.14 is estimated for the entire period 1760–1800.

- Real GDP growth <2% until mid-19th century
- Population growth ~1.4% per year
- TFP growth ~0.7% (not spectacular)
- By 1851: more in services than in industry
- Britain's global industrial share rose dramatically

THE SLOW GROWTH PARADOX

	Rates of growth (per thousand per annum)			
	<i>1600–1700</i>	<i>1700–50</i>	<i>1750–1800</i>	<i>1800–50</i>
France	1.4	1.7	3.5	4.3
Germany		1.2	7.3	7.4
Italy	0.1	3.0	2.9	6.0
The Netherlands	2.4	0.0	2.0	7.8
Spain	1.0	3.0	4.2	6.7
Sweden		5.3	5.6	7.9
England	2.5	2.6	7.5	13.6

THE SLOW GROWTH PARADOX

1. Rapid **demographic expansion**
 2. Technological innovation initially affected only a **limited** range of **sectors**.
 3. The sectors impacted at the outset were relatively **small in scale**.
 4. New technologies were highly **specific in scope**:
 - they applied to only one stage of the production process
 - the steam engine represented the sole 'general purpose' technology (GPT)
 5. The transition from invention to innovative products and processes does not occur overnight.
 - The relationship between science and technology is weak:
 - new technologies are imperfect
 - The supply of adequate skills in the labor market is initially limited.
 6. The efficiency of the factory system follows learning curves.
 - Entrepreneurs → must learn how to manage the workforce
 - Workers → must adapt to factory discipline.
- THE PROCESS IS GRADUAL IN SECTORS,
INNOVATION, ADAPTATION, PROGRESS!

HOW CAN THIS BE OVERCOME?

INVENTION



- ☐ New idea, new scientific development, new product, new process
- ☐ Expands the existing stock of technological knowledge
- ☐ Can occur anywhere; random
- ☐ Often the result of individual action – does not necessarily imply innovation

INNOVATION

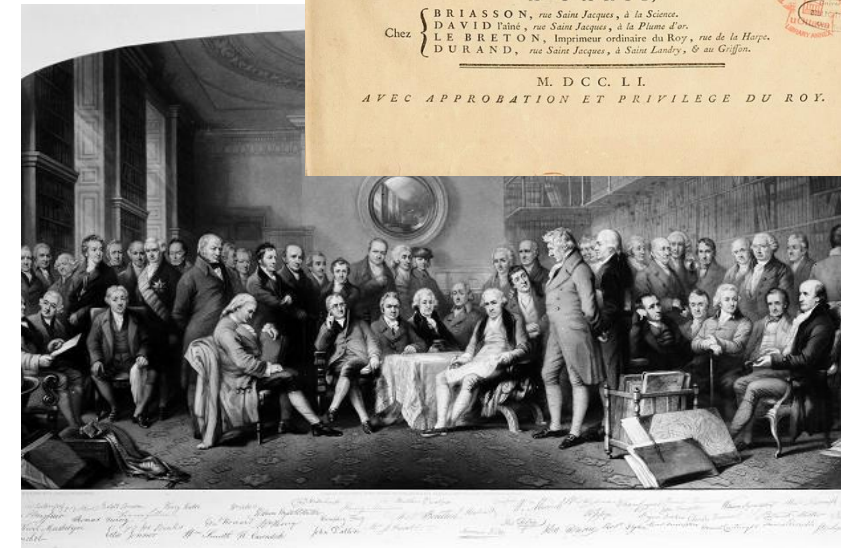
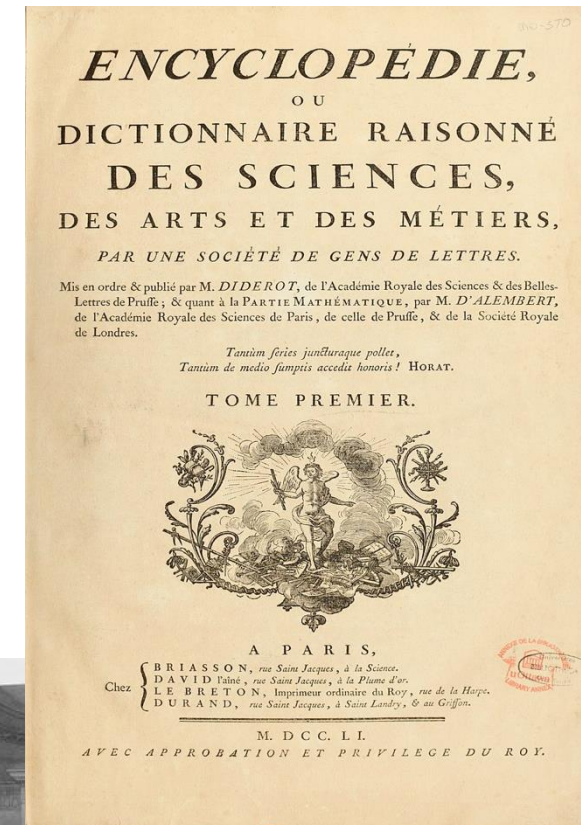


- ☐ Introduction into the use and commercial exploitation of the invention
- ☐ Requires articulated knowledge: markets, production, resources, technology
- ☐ Requires interaction with other individuals
- ☐ Conditioned by markets and institutions
- ☐ Has an economic and social character

DOES IT MEAN MICRO I. HAVE BEEN
USELESS?

DOES CULTURE MATTER? THE EUROPEAN ENLIGHTENMENT

- Scientific Revolution: the origins
 - *Francis Bacon*: understanding the world through scientific inquiry
 - *Isaac Newton*: science can establish the laws of nature
- The Enlightenment
 - By 18th century, scientific approach accepted in western Europe
 - Enlightened governments established institutions to promote science (academies, societies, patent system)
- **Mokyr**: Enlightenment caused industrialisation
 1. New scientific culture increased the rate of innovation
 2. Institutions + high literacy → lower cost of access to knowledge
 3. Intellectual property rights guaranteed by patents

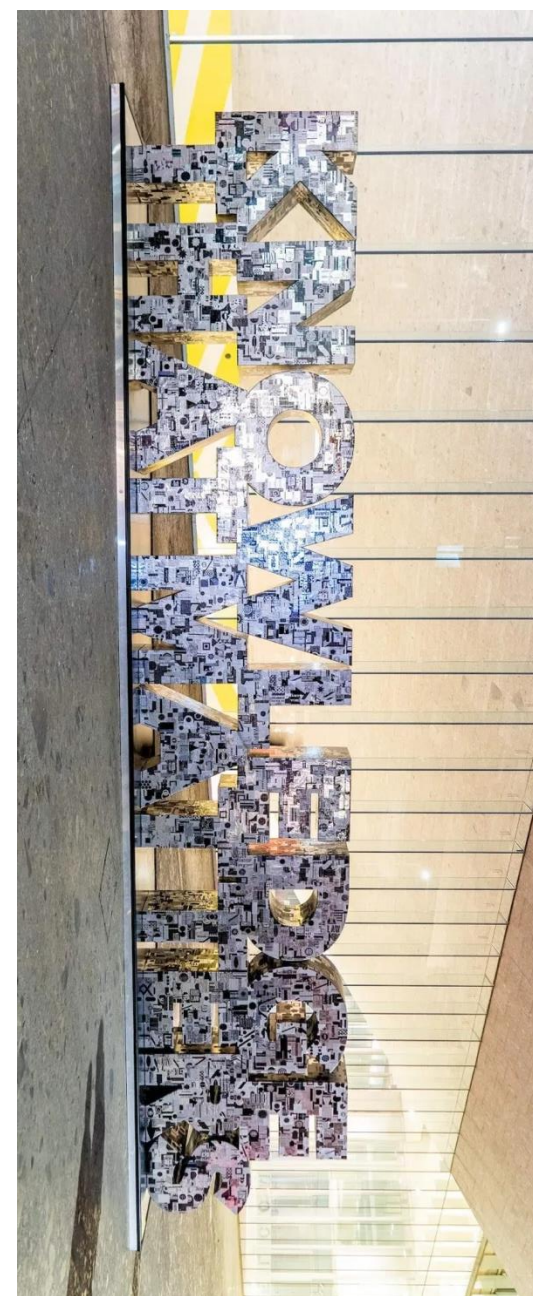


THE EUROPEAN ENLIGHTENMENT

- The Industrial Revolution
 - Flow of new inventions mobilized new investment
 - ➔ higher labour productivity ➔ higher wages ➔ labour-saving innovation
- Disagreement with Allen
 - Macro inventions: did not bring much benefit in 18th century
 - Micro inventions: improved science in the 19th century made industrial technology more efficient
 - Productivity growth mostly came from micro inventions after 1820.
- **Mokyr**: enlightened culture and institutions created an environment supportive of micro inventions

THE EUROPEAN ENLIGHTENMENT

- Origins of modern economic growth
 - The Industrial Revolution was not the first major cluster of inventions, but the first that turned into continuous technological progress.
- Epistemic base
 - Pre-industrial technology lacked epistemic base; trial and error
 - Micro-inventions improve on existing technologies → inventors need to understand why and how these technologies work
- Useful knowledge
 - Scientific inquiry is useful if it has practical value and solves problems.
 - Growth of useful knowledge must be a collective enterprise as its gains are broadly shared.



THE EUROPEAN ENLIGHTENMENT: TWO SIDES OF THE SAME MEDAL

1. **Agenda** – focus on practical, useful problems.
 2. **Capabilities** – better instruments, math, research methods.
 3. **Selection** – tolerance & persuasion in the “market for ideas.”
 4. **Diffusion** – open science, encyclopedias, public lectures
- Inventing is a risky business
 - Opportunity cost: time investment
 - Unpredictable returns: investment risk
 - Knowledge is non-excludable
 - Once diffused, anyone can have it for free
 - Solution: intellectual property rights
 - Patents
 - Copyright

COMPLEMENTARITY OF MICRO AND MACRO INVENTION

Industrial Revolution: 1st stage (1760s-1820s)

- ▶ Technological progress limited to few industries; the epistemic base was insufficient for their broader application.
- ▶ Micro-inventions were slow and often unsuccessful, leading to **modest productivity growth** in Britain before 1820.

Industrial Revolution: 2nd stage (1830s-1870s)

- ▶ Full mechanisation of textile spinning and weaving; improvement in iron and steel furnaces; precision engineering; gas lighting
- ▶ High-pressure steam engines revolutionised transportation; telegraph and industrial printing did the same in communication.
- ▶ Scientific breakthroughs enabled the harnessing of electricity and industrial chemistry (fertilisers, chemical dye-stuffs, etc.)
- ▶ **Fast and sustained productivity growth.**

TECHNOLOGY IN THE TEXTILE

Table 8.1 *Real cost of cotton (16 count) in 1784 prices (d/lb)*

	Hand method, 1760	24-spindle jenny, 1775	Arkwright mill, 1784	Glasgow (integrated) mill, 1836
Raw cotton	16.88	16.88	16.88	16.70
Preparatory	1.40	1.40	0.82	0.02
Carding	5.60	5.60	1.87	0.14
Spinning	7.00	2.33	2.57	0.34
Reeling, bundling	0.47	0.47	1.07	0
Craft and other			1.12	
Administrative	2.72	2.72	0.41	0.02
Materials	0	0	1.20	0.53
Capital	0.93	1.88	2.00	0.47
Average total cost	35.00	31.28	27.94	18.22

The figures in the table are of varying degrees of reliability.

These new innovations were fundamental for the development of the textile industry:

- wool
- silk
- cotton

However, the factory system was not yet achieved. It was only with the introduction of Arkwright's hydraulic spinning machine (1769), which was much heavier and more expensive than the previous spinning machines, that we have the development of the factory system.

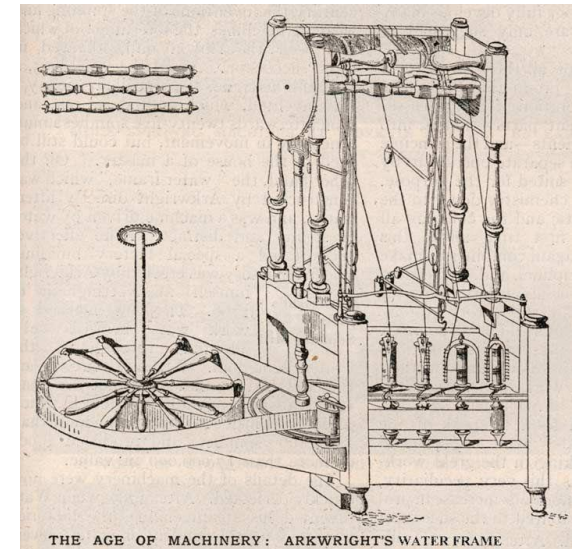
As large quantities of water were needed, the first factories were built in the countryside and in small towns.



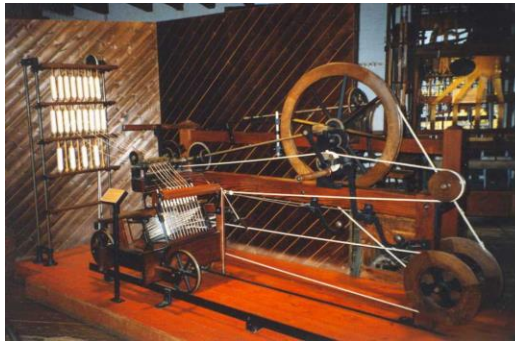
The medieval spinning wheel



1764 - Spinning Jenny

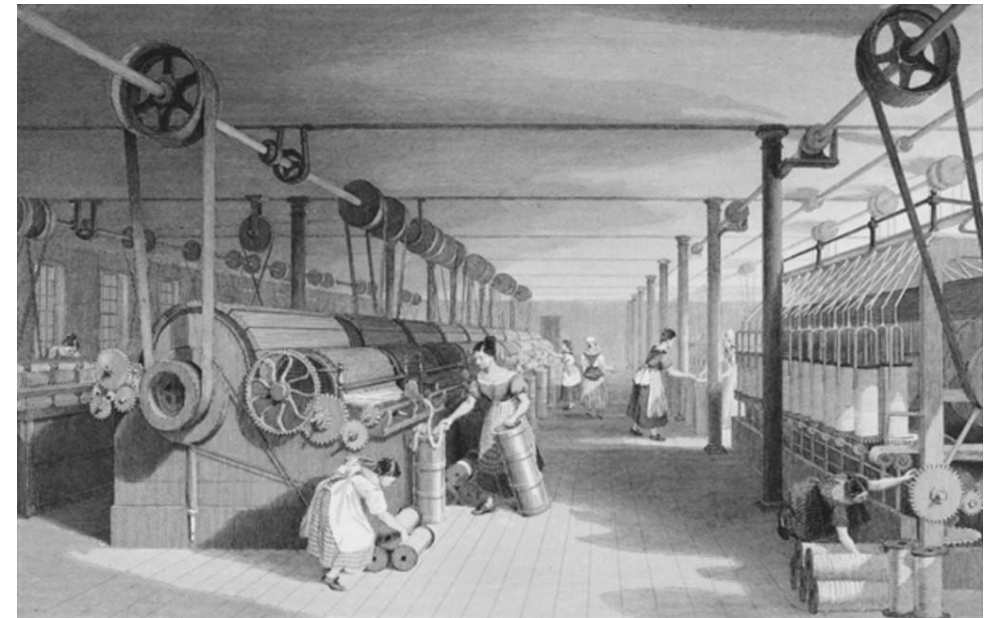


1769 - Water frame



1774-1779 - Mule

**THE NAISSANCE OF
THE FACTORY:
A REVOLUTION
WITHIN THE
REVOLUTION!!!!**



THE COTTON INDUSTRY

- Before 1780, Britain was a minor producer of cotton cloth. By 1830, cotton textiles were the largest British industry:
 - 8% of Gross Domestic Product
 - 16% of manufacturing jobs
- Britain's early technological lead secured a competitive advantage throughout the 19th century.

Tons of raw cotton processed in cotton spinning by the industrial powers per year

	UK	France	Germany	USA
1840s	192,000	54,000	11,000	47,000
1900s	869,000	231,000	435,000	1,110,000

THE COTTON DISTRICT: LANCASHIRE

- ▶ Lancashire became a dominant hub for the cotton industry during the Industrial Revolution due to a combination of favorable geography, economic conditions, and acquired advantages that created a powerful industrial agglomeration.
- ▶ Lancashire had "first nature geography" that provided crucial resources.



Water and power: the region's fast-flowing streams and rivers supplied the necessary water power for early mills, a vital resource before the widespread adoption of steam power.

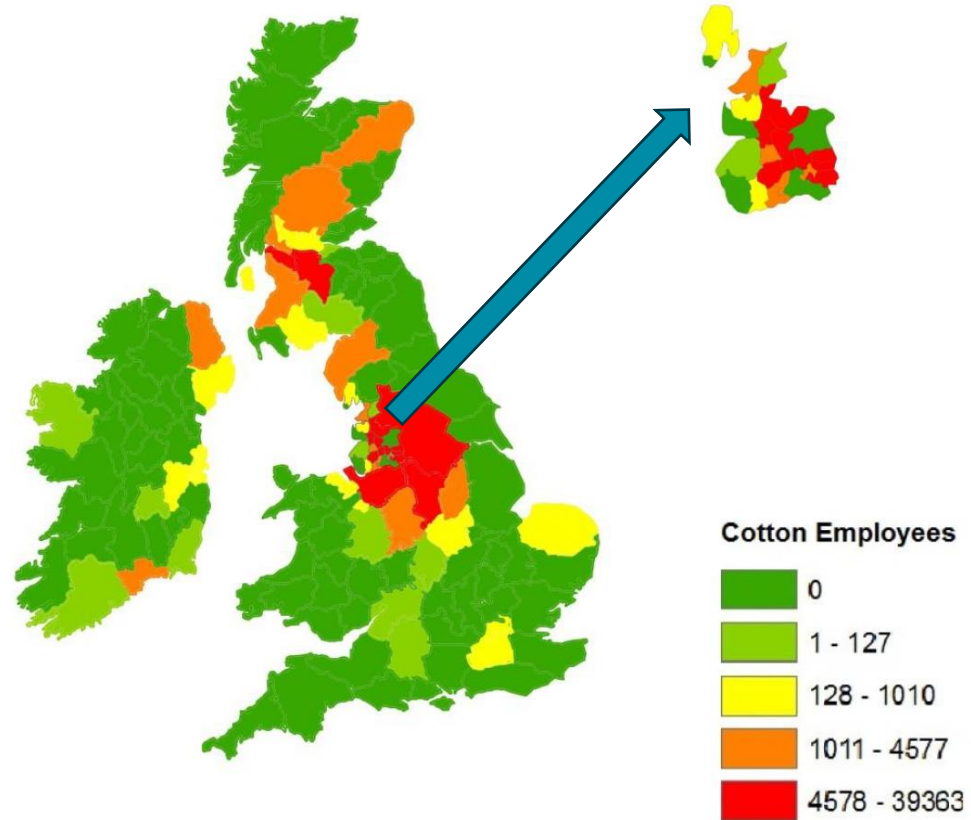


Coal: the availability of cheap local coal was a key advantage, especially as steam power replaced water power. This allowed the industry to continue its growth and move away from being solely dependent on rivers.



Access to the sea: Lancashire had excellent access to the sea, particularly through the port of Liverpool, which was essential for importing raw cotton and exporting finished goods.

Figure 1. The location of Employment in the Cotton Industry in Britain 1838



THE COTTON DISTRICT: LANCASHIRE

- ▶ Lancashire became an extremely successful industrial agglomeration.
- ▶ This concentration of related businesses led to significant productivity benefits.
 - ▶ The dense network of suppliers, technological spillovers, and a large pool of labour with specialized skills made the area highly productive.
- ▶ This high productivity allowed companies to pay higher wages while remaining competitive in international markets.

Table 4: British cotton textile exports in the Indian market, 1831-35 to 1880-81

	Indian consumption of cotton textiles (m lb)	Share taken by British exports (%)
1831-35	375	3.9
1856-60	455	35.3
1880-81	600	58.4

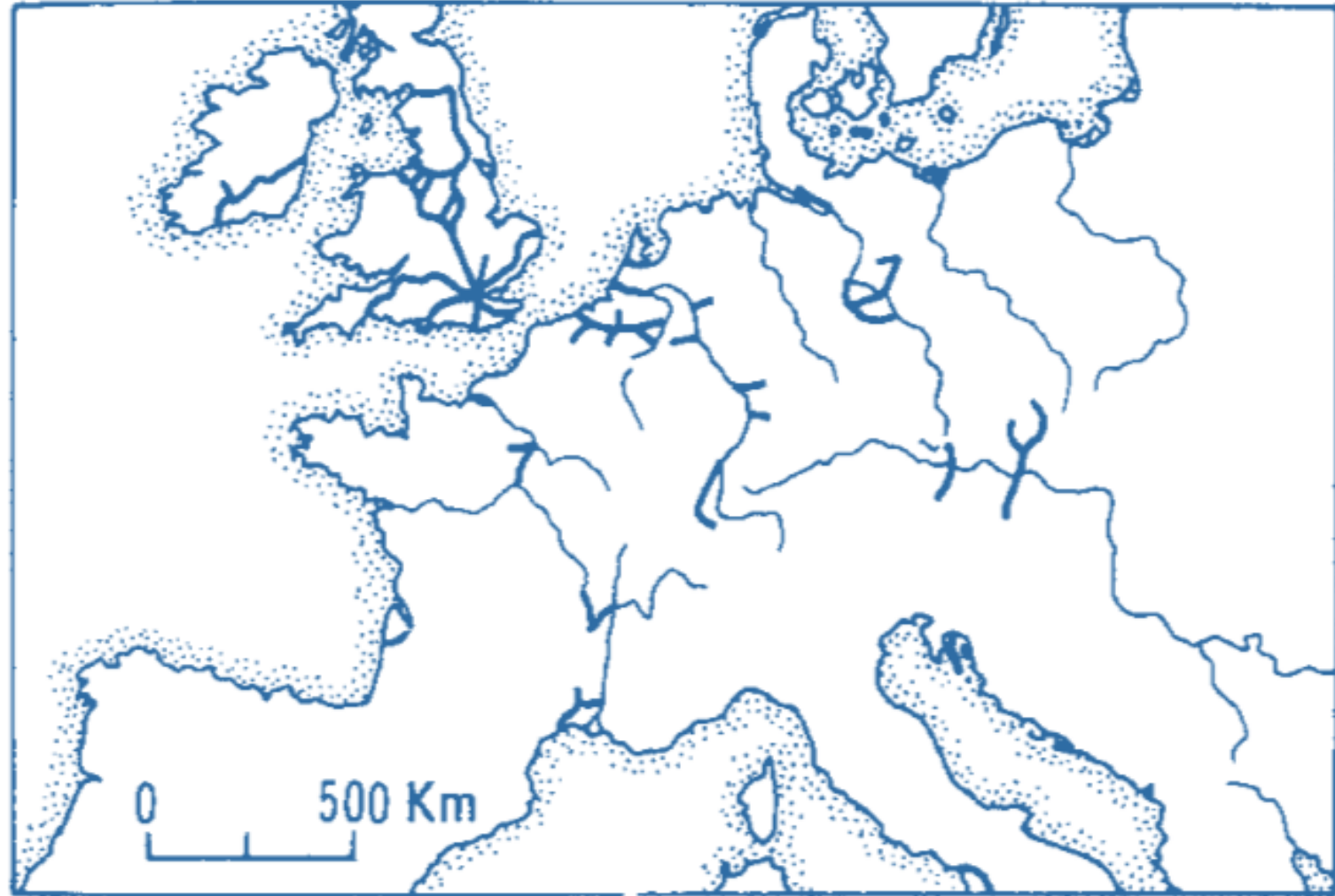
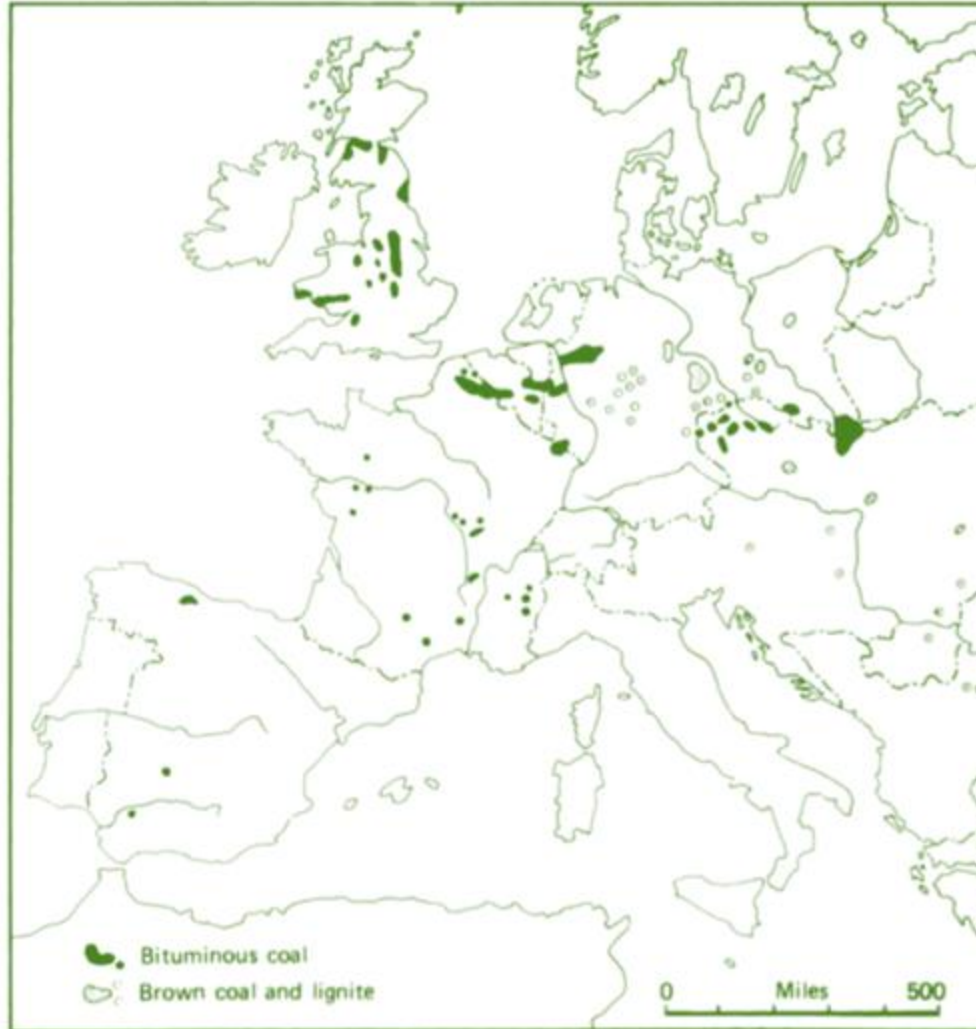
By 1851, the Lancashire industry accounted for 66% of UK spindles and about **46% of the world's spindles**.

THE MINING INDUSTRY

- ▶ The innovations that contributed most to English industrial success are:
 - ▶ the process of smelting ferrous metal using coke
 - ▶ the invention of the steam engine
- ▶ **Coke** is obtained by bringing the coal at very high temperatures to remove impurities in the form of gas.
- ▶ It is a fuel with a high calorific value that made it easier to process iron and pig iron.

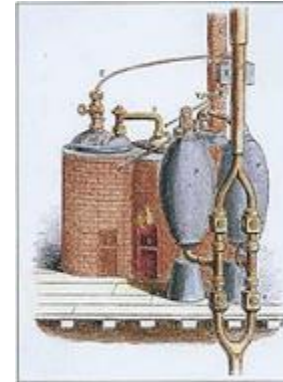


FIND THE DIFFERENCES

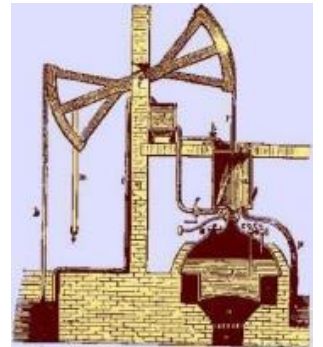


THE STEAM POWER

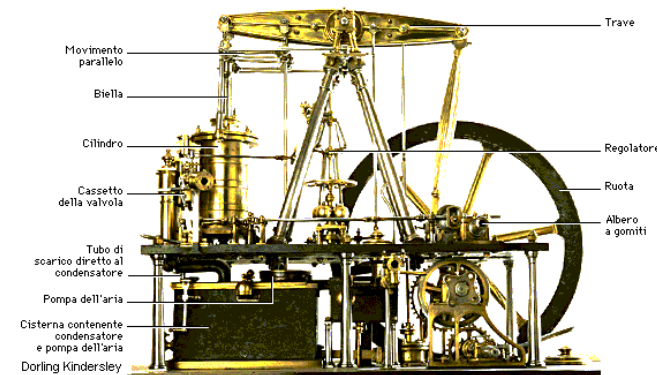
- ▶ The higher demand for iron and pig iron (crude iron) led to the expansion of the mining industry and the search for better techniques to extract these minerals.
- ▶ The mines became deeper and deeper and this forced technicians and engineers to find a solution to free the galleries from water: → hydraulic pumps driven by steam engines were introduced.
- ▶ Key innovations of this period include James Watt's invention of a more energy-efficient steam engine, and one capable of transforming piston motion into rotary motion.



1698 Thomas Savery



1712 Thomas Newcomer



1771 James Watt

WHY BRITAIN?

- Market economy with property rights & rule of law + Strong but limited state, sound taxation
→ HIGHLY OPEN LIBERAL ECONOMY, FREE TRADE BIAS
- Favourable geography: coal, water, sea access
- Skills in textiles, mining, iron,
→ COMPLEMENTARITY AMONG SECTORS
- Agriculture reorganized on capitalist lines



CONSOLIDATING THE LEAD: THE 'WORKSHOP OF THE WORLD'

- Cotton textiles as iconic sector
→ KNOWLEDGE
- Lancashire as industrial hub (agglomeration effects)
→ SPECIALIZATION
- Agglomerations → productivity, hard to replicate
→ ECONOMIES OF SCALE
- Exports: ~25% of industrial output by 1851
→ EXPORT ORIENTED